

Your Gut & Parkinson's Disease

What the Research Says — In Plain Language

You may already know that Parkinson's affects movement — but did you know it also has a deep connection to your gut? Researchers are discovering that the health of your digestive system plays a much bigger role in Parkinson's than anyone realized. Here is what the science says, in plain language.

■ The Gut–Brain Connection: Why Your Gut Matters

Your gut and your brain are in constant communication through a pathway researchers call the **gut–brain axis**. Think of it as a two-way highway between your digestive system and your brain, sending signals back and forth all day long.

In Parkinson's, this communication system can be disrupted. Research has found that people with Parkinson's often have changes in their **gut microbiome** — the trillions of bacteria that live in your digestive tract. These changes can affect inflammation, brain health, and many of the non-motor symptoms you may experience.

“The gut microbiota is strongly associated with Parkinson's disease. Intestinal issues often occur before motor symptoms appear — suggesting the gut may play a role early in the disease process.” — Translational Neurodegeneration, 2023

One protein central to Parkinson's — called **alpha-synuclein** — is believed to potentially begin accumulating in the gut and travel to the brain via the vagus nerve (the main nerve connecting gut and brain). This is an active and exciting area of research.

■ Constipation: A Common Challenge in Parkinson's

Constipation is one of the most common non-motor symptoms in Parkinson's, and it is often under-discussed. Research shows that gut movement (motility) is frequently slowed in people with Parkinson's, partly because dopamine — the same brain chemical affected by PD — also plays a role in regulating the digestive system.

Good news: Studies show that dietary changes can meaningfully improve constipation in people with Parkinson's — sometimes within just a few weeks.

■ Study #1: Mediterranean Diet — 5 Weeks to Fewer Symptoms

What researchers did:

Scientists at the University of Florida followed people with Parkinson's who followed a Mediterranean diet for just 5 weeks. They measured constipation scores, gut bacteria composition, and other digestive markers before and after.

What they found:

- Constipation scores **significantly improved** after 5 weeks
- Gut bacteria changed in helpful ways — beneficial bacteria increased, while potentially harmful strains decreased
- Participants also experienced **weight loss**
- The diet was shown to be **realistic and achievable** for people with PD

Key takeaway: Just 5 weeks of eating a Mediterranean diet made a measurable difference in gut health and constipation for people living with Parkinson's.

Source: Rusch et al., *Frontiers in Neurology*, December 2021 — University of Florida, Norman Fixel Institute for Neurological Diseases

■ Study #2: A Larger, Stronger Study Confirms the Results (2024)

Because the original 2021 study was small, the same research team returned with a larger, more rigorous **randomized controlled trial** — the gold standard of medical research. Published in September 2024, it compared people following a Mediterranean diet against a standard care group over 8 weeks.

What they found:

- Both groups saw reduced constipation — but the Mediterranean diet group consumed **significantly more dietary fibre**
- The Mediterranean diet group showed **lower intestinal inflammation** (measured by a marker called calprotectin)
- This suggests the Mediterranean diet does more than just relieve symptoms — it may actually **reduce inflammation in the gut**

Why does gut inflammation matter? Chronic inflammation in the gut is thought to contribute to Parkinson's disease progression. Reducing it through diet is a simple, safe, and powerful tool.

Source: Rusch et al., *Nutrients*, September 2024 — University of Florida, Norman Fixel Institute for Neurological Diseases

■ Study #3: A Review of All the Evidence (2024)

A 2024 review from University College London looked at all available studies on Mediterranean diet, gut microbiome, and Parkinson's together. Reviewing both observational studies (over 1,200 participants) and clinical trials, they found consistent patterns across the research.

Consistent findings across studies:

- High Mediterranean diet adherence was linked to beneficial changes in gut bacteria — especially an increase in bacteria that produce **short-chain fatty acids** (which protect the gut lining)
- Improvements were seen in **constipation, cognitive function, and indigestion**
- No significant effect on motor symptoms was found — but researchers note more and longer studies are needed

Source: Seelarbokus et al., *Nutrients*, July 2024 — University College London, UCL Queen Square Institute of Neurology

■ What Does the Mediterranean Diet Actually Look Like?

The good news: this is not a restrictive or complicated diet. It focuses on whole, natural foods that are colourful, flavourful, and satisfying.

Eat Plenty Of	Limit or Reduce
• Vegetables — as many colours as possible • Fresh fruit • Whole grains (oats, brown rice, quinoa) • Legumes (lentils, chickpeas, beans) • Nuts and seeds • Olive oil as your main fat • Fish, especially fatty fish (salmon, sardines) • Herbs and spices for flavour	• Ultra-processed foods • Fast food and fried foods • Sugary drinks and sweets • Red meat (occasional is fine) • Highly refined white bread/pasta • Packaged snack foods

■ Other Promising Research Areas

Beyond diet, researchers are exploring other ways to support gut health in Parkinson's. While these are still emerging, the early findings are encouraging:

- **Probiotics and prebiotics** — beneficial bacteria supplements may help improve bowel motility and reduce gut inflammation
- **Resistant starch** — a type of fibre found in foods like cooled potatoes, unripe bananas, and oats has been studied in a clinical trial (RESISTA-PD) for its effects on gut bacteria in Parkinson's
- **Fecal microbiome transplant (FMT)** — transferring healthy gut bacteria from a donor into a patient's gut; early studies show promise for constipation relief, though this is still experimental

Bottom line: Diet remains the most practical, safe, and evidence-supported tool available right now for supporting gut health in Parkinson's.

■ What You Can Start Doing Today

- Add one more vegetable to your plate at lunch and dinner
- Swap refined bread or pasta for a whole grain version
- Add a handful of nuts or seeds as a daily snack
- Use olive oil instead of butter when cooking
- Aim for fish twice a week — salmon, sardines, or trout are great choices
- Drink water consistently throughout the day to support bowel regularity
- Add legumes (lentil soup, chickpea salad) to your weekly meals

You do not need to overhaul your entire diet overnight. Small, consistent changes add up. Research shows that even short-term improvement in eating patterns can produce meaningful changes in your gut health.

The connection between your gut and your brain is real, and the research is growing. What you eat is one of the most powerful tools you have to support your health with Parkinson's — alongside exercise, sleep, stress management, and community. You are not powerless. Every meal is an opportunity. Questions about nutrition and your Parkinson's Pathway to Empowerment program? Reach out to Christine at info@boxing4health.com or 613.224.2694.